

Calculator Trust Test Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Medium	ID	calculator-trust-test

Big idea: Tools need interpretation.

Overview

Students investigate how mathematical technology can both reveal and mislead.

Student Prompt

Inspect Calculator Trust Test Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Calculator Trust Test Stimulus A	Student-facing stimulus 1: A calculator output checked against estimation, mental maths, and another tool, such as $\log_{10}(1000) = 3$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Calculator Trust Test Stimulus B	Student-facing stimulus 2: A deliberately mistyped expression that creates a plausible wrong answer. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Calculator Trust Test Reveal / Twist	Student-facing stimulus 3: A trust checklist for tools: input, process, output, and independent check. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Tools need interpretation.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Calculator Trust Test Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Calculator Trust Test Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Does technology make mathematical knowledge more reliable?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how tools need interpretation.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare a starter stimulus using: A calculator output checked against estimation, mental maths, and another tool, such as $\log_{10}(1000) = 3$.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- A calculator output checked against estimation, mental maths, and another tool, such as $\log_{10}(1000) = 3$.
- A deliberately mistyped expression that creates a plausible wrong answer.
- A trust checklist for tools: input, process, output, and independent check.

Suggested Timing

Stage	Teacher move	Watch for
1	Give examples where technology can mislead: restricted graph window, rounded decimals, regression on inappropriate data, or a graph that hides an asymptote.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students explain what the tool displays and what it does not display.	Assumptions, confidence shifts, evidence claims, and language choices.
3	They write a warning label for the tool.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Discuss responsibility for interpretation.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.

- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Proof versus Pattern
- Monty Hall Shock
- Axiom Switch Universe